

Exam Score: _____

Name: Answers

1. Suppose that $A = \begin{bmatrix} 1 & 0 & 2 & 0 & -1 \\ 2 & -1 & 3 & -2 & -2 \\ 0 & 2 & 2 & 4 & 0 \\ -2 & 0 & -4 & 1 & 4 \end{bmatrix} \xrightarrow{\text{row operations}} B = \begin{bmatrix} 1 & 0 & 2 & 0 & -1 \\ 0 & 1 & 1 & 0 & -4 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

a) Fill in the following blanks:

(6) (i) $\text{rank}(A) = 3$ (ii) $\dim(\text{Nul}(A)) = 2$ (iii) $\dim(\text{Nul}(A^T)) = 1$

b) Give a basis for the null space of A .

$x_1 + 2x_3 - x_5 = 0$
 $x_2 + x_3 - 4x_5 = 0$
 $x_4 + 2x_5 = 0$
 $x_3, x_5 \in \mathbb{R}$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -2x_3 + x_5 \\ -x_3 + 4x_5 \\ x_3 \\ -2x_5 \\ x_5 \end{bmatrix} = x_3 \begin{bmatrix} -2 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 1 \\ 4 \\ 0 \\ 2 \\ 1 \end{bmatrix}$$

(10) $x_3, x_5 \in \mathbb{R}$

A basis is given by $\begin{bmatrix} -2 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 0 \\ 2 \\ 1 \end{bmatrix}$

c) Give a basis for the row space of A .

$[1 \ 0 \ 2 \ 0 \ -1], [0 \ 1 \ 1 \ 0 \ -4], [0 \ 0 \ 0 \ 1 \ 2]$

(6)

d) Give a basis for the column space of A .

(6) $\begin{bmatrix} 1 \\ 2 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 4 \\ 1 \end{bmatrix}$

2. For the following parts, let A be a 3×3 matrix. Suppose that

$$A \xrightarrow{2R_1+R_2 \rightarrow R_2} B \xrightarrow{R_2 \leftrightarrow R_3} C \xrightarrow{-3R_2+R_3 \rightarrow R_3} D \xrightarrow{-R_3+R_1 \rightarrow R_1} I_3$$

a) Is A invertible? If so, then find its inverse and if not explain why not.

$$\begin{aligned} I_3 &\xrightarrow{2R_1+R_2 \rightarrow R_2} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} \xrightarrow{-3R_2+R_3 \rightarrow R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 2 & 1 & -3 \end{bmatrix} \\ &\xrightarrow{-R_3+R_1 \rightarrow R_1} \begin{bmatrix} -1 & -1 & 3 \\ 0 & 0 & 1 \\ 2 & 1 & -3 \end{bmatrix} = A^{-1} \end{aligned}$$

$$A = \begin{bmatrix} -1 & -1 & 3 \\ 0 & 0 & 1 \\ 2 & 1 & -3 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 1 \\ -2 & 3 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

b) Give the 3×3 matrix M such that $MD = A$.

$$M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 2 & 1 & -3 \end{bmatrix}^{-1} =$$

Alternatively

$$I_3 \xrightarrow{3R_2+R_3 \rightarrow R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 1 & 0 \end{bmatrix} \xrightarrow{-2R_1+R_2 \rightarrow R_2} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 3 & 1 \\ 0 & 1 & 0 \end{bmatrix} = M$$

3. For the following parts let $A^{-1} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, and let $B^{-1} = \begin{bmatrix} -2 & 3 \\ -1 & 2 \end{bmatrix}$.

a) Circle the matrix that equals $(AB)^{-1}$

(i) $\begin{bmatrix} -4 & 7 \\ -10 & 17 \end{bmatrix}$

(ii) $\begin{bmatrix} 17/2 & -7/2 \\ 5 & -2 \end{bmatrix}$

(iii) $\begin{bmatrix} 3 & -4 \\ -7/2 & 7/2 \end{bmatrix}$

(iv) $\begin{bmatrix} 7 & 8 \\ 5 & 6 \end{bmatrix}$

b) Circle that solution for the system $Ax = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$

(i) $\begin{bmatrix} 7 \\ -9/2 \end{bmatrix}$

(ii) $\begin{bmatrix} 4 \\ 6 \end{bmatrix}$

(iii) $\begin{bmatrix} 4 \\ 18 \end{bmatrix}$

(iv) $\begin{bmatrix} 13 \\ 8 \end{bmatrix}$

4. (pt) Find the unit lower triangular matrix $L \in \mathbb{R}^{3 \times 3}$ such that $A = LU$ where A and U are given in the following:

$$A = \begin{bmatrix} 1 & 0 & 1 & 3 \\ 0 & 2 & 1 & 1 \\ 3 & -4 & 1 & 7 \end{bmatrix} \xrightarrow{-3R_1 + R_3 \rightarrow R_3} \begin{bmatrix} 1 & 0 & 1 & 3 \\ 0 & 2 & 1 & 1 \\ 0 & -4 & -2 & -2 \end{bmatrix} \xrightarrow{2R_2 + R_3 \rightarrow R_3} U = \begin{bmatrix} 1 & 0 & 1 & 3 \\ 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} \quad \downarrow \quad \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix}$$

Alternatively:

$$I_3 \xrightarrow{-2R_1 + R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \xrightarrow{3R_1 + R_3 \rightarrow R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix} = L$$

5. (pt) Let $B = \{v_1, v_2, v_3\}$ be the basis for $\mathbb{R}^3$ given by $v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$, $v_3 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$. Given

(12) $u = \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix}$, find the coordinates for u with respect to the basis B , i.e. $[u]_B$.

$$[u]_B = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \text{ where } [v_1 \ v_2 \ v_3] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \\ 1 & 2 & 0 & 4 \end{bmatrix} \xrightarrow{\text{row ops}} \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$[u]_B = \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix}$$

6. For the following parts, let $A = \begin{bmatrix} h & 2 & -1 \\ 1 & h & 2 \\ 0 & 1 & 1 \end{bmatrix}$.

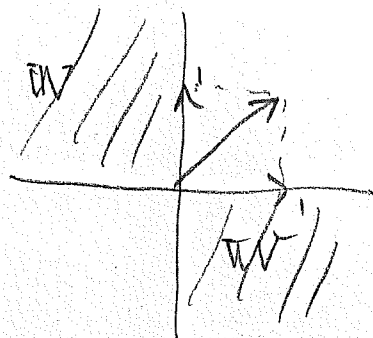
a) Use cofactor expansion along the first column to compute $\det A$

$$\textcircled{10} \quad h \begin{vmatrix} h & 2 \\ 1 & 1 \end{vmatrix} - 1 \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix} = h(h-2) - (2+1) = h^2 - 2h - 3 \\ = \underline{(h-3)(h+1)}$$

b) Give all the values of h for which A is *not* invertible.

$$\textcircled{6} \quad \text{Roots of } (h-3)(h+1), \text{ i.e. } \underline{\underline{h=3, -1}}$$

7. Explain why the set $W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \mid xy \leq 0 \right\}$, is not a subspace of $\mathbb{R}^2$. Note that the set W consists of the second and fourth quadrants of $\mathbb{R}^2$, including the axes.



$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \in W \quad \text{but} \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \notin W$$

So W is not closed under vector addition